EC571: Digital VLSI Circuit Design

Homework 5 — Fall 2026

Total: 100 points

Due: November 7, 2026

General Instructions

Use this worksheet to complete the homework. Please print this worksheet, write your solutions on the worksheet, take a picture of the worksheet, and submit it through Gradescope. If you prefer, you can download the Homework 5 worksheet on your tablet, complete the worksheet on your tablet, and submit the completed worksheet through Gradescope.

Show all reasoning. A final numerical answer without intermediate steps will receive limited credit.

There are five questions, each worth 20 points. Answer all questions and show your calculations and reasoning.

For timing problems, clearly distinguish between maximum-delay (setup) and minimum-delay (hold) constraints. State any additional assumptions you make.

You may use course notes, lecture slides, and a calculator. AI tools may be used for concept review but not to generate final solutions. Include a brief AI-use disclosure with your submission.

Write your responses in the blank space provided under each part; attach additional pages through Gradescope if necessary.

Question 1: Setup/Hold Characterization and Register Selection [20 points]

A circuit designer is evaluating two positive-edge-triggered registers, Register A and Register B, for use in a timing-critical pipeline.

Register A has been characterized by changing the arrival time of its input data relative to the rising clock edge and measuring the resulting clock-to-Q delay.

The following data was obtained:

Time before clock edge at which D changes (ps)	Clock-to-Q delay (ps)
160	70
110	70
90	71
80	73
60	77
45	93

To characterize the hold time, the designer keeps the correct input data stable through the rising clock edge and then changes the input at different times after the edge.

Time after rising clock edge (ps)	Observed register behavior
10	Incorrect value captured
25	Output glitch observed
40	Correct output, no glitch
60	Correct output, no glitch

Use a 5% increase in clock-to-Q delay relative to its nominal value as the criterion for defining setup time. Linear interpolation between characterized points is permitted.

(a) **Setup-time extraction.** **[6 points]**

Estimate the setup time of Register A using the characterization data. Explain why the clock-to-Q delay increases as the input transition approaches the active clock edge.

(b) **Hold-time extraction.** **[4 points]**

Determine a conservative hold-time requirement for Register A based on the observations. Explain how a hold-time violation differs from a setup-time violation in terms of what happens inside a master-slave register.

(c) **Register comparison.** **[6 points]**

Register B has the following characteristics:

Parameter	Register A	Register B
Maximum clock-to-Q delay	70 ps	82 ps
Minimum clock-to-Q delay	38 ps	45 ps
Setup time	From part (a)	50 ps
Hold time	From part (b)	55 ps
Clock input capacitance	4 fF	7 fF

Two identical registers are connected by a combinational circuit with a maximum delay of 360 ps and a minimum delay of 20 ps.

Assume a clock period of 500 ps, zero clock skew, and no clock uncertainty.

For both register choices, determine the setup and hold timing margins. Which design meets both timing constraints?

(d) **Design recommendation.** **[4 points]**

Suppose the goal is to operate at the highest possible clock frequency while minimizing clock-distribution power.

Which register would you select? Discuss the tradeoff between clock-to-Q delay, setup time, hold time, and clock input capacitance.

Question 2: Clock Skew, Jitter, and Timing Closure [20 points]

A digital system contains two positive-edge-triggered registers, R1 and R2, connected through two alternative combinational paths. Depending on the input combination, either path may be sensitized, so both must satisfy timing requirements.

The circuit operates with a clock period of 620 ps.

The timing parameters are:

Parameter	Value
Maximum clock-to-Q delay of R1	80 ps
Minimum clock-to-Q delay of R1	35 ps
Setup time of R2	65 ps
Hold time of R2	55 ps
Setup clock uncertainty	15 ps
Hold clock uncertainty	10 ps

The two combinational paths have the following delays:

Path	Minimum delay	Maximum delay
P1	100 ps	470 ps
P2	15 ps	320 ps

Clock skew is defined as the arrival time of the clock at R2 minus the arrival time at R1. Positive skew means that the capture clock arrives later than the launch clock.

Assume the setup and hold uncertainties reduce their corresponding timing margins by the specified amounts. All propagation delays are measured at the appropriate process and operating corners.

(a) **Timing without skew.** [4 points]

Assume zero clock skew.

Calculate the worst-case setup and hold slacks considering both combinational paths. Identify the critical path for each timing constraint.

(b) **Effect of clock skew.** **[6 points]**

Now consider two clock-distribution implementations:

- Design A: +20 ps clock skew
- Design B: −20 ps clock skew

Compute the setup and hold slacks for both designs.

Which implementation is better for setup timing? Which is better for hold timing? Explain why clock skew affects the two constraints differently.

(c) **Feasible skew range.** **[4 points]**

The clock-tree designer can intentionally adjust the clock skew anywhere between −40 ps and +40 ps.

Determine whether any skew value can satisfy both setup and hold constraints simultaneously without modifying the combinational logic.

Justify your answer.

(d) **Timing repair.** **[6 points]**

A delay buffer is added to path P2. The buffer increases both the minimum and maximum delays of P2 by 30 ps, without affecting P1.

Determine whether there is now a feasible clock-skew range that satisfies both setup and hold timing.

Select a skew value within that range, calculate the resulting timing margins, and explain why the timing repair is effective.

Would the same strategy necessarily be effective if P2 were also the longest combinational path? Explain.

Question 3: Pipeline Partitioning and Minimum Clock Period [20 points]

A digital processing block performs four operations in sequence:

$$\mathbf{A} \longrightarrow \mathbf{B} \longrightarrow \mathbf{C} \longrightarrow \mathbf{D}$$

The block is initially implemented as a single combinational path between a launch register and a capture register.

The operation delays are:

Logic block	Minimum delay (ps)	Maximum delay (ps)
A	30	135
B	40	270
C	25	160
D	35	205

All registers have the following timing parameters:

Parameter	Value
Maximum clock-to-Q delay	65 ps
Minimum clock-to-Q delay	35 ps
Setup time	55 ps
Hold time	45 ps

Clock skew can vary between -25 ps and $+25$ ps. Assume a worst-case setup uncertainty of 20 ps and a worst-case hold uncertainty of 10 ps. Use the adverse skew polarity independently for setup and hold checks.

The designer can insert up to two additional pipeline registers, but only at the boundaries between A, B, C, and D. Ignore the delay of interconnect between adjacent logic blocks.

(a) **Original design.** [4 points]

Calculate the minimum clock period of the original unpipelined design, accounting for the worst-case setup conditions.

(b) **Optimal pipeline partition.** **[6 points]**

Determine where to place two additional pipeline registers to minimize the clock period.

Evaluate all possible placements of the two registers and identify the best partition.

For your chosen partition, calculate the minimum clock period and maximum operating frequency.

(c) **Throughput versus latency.** **[5 points]**

Suppose the designer is permitted to insert only one additional pipeline register.

Determine its best location and calculate the corresponding minimum clock period.

Compare the original design, the one-additional-register design, and the two-additional-register design in terms of:

- Maximum operating frequency
- Steady-state throughput, assuming one result can be produced per clock cycle after the pipeline fills
- Number of clock cycles required for a data item to travel from the original launch register to the final capture register

Explain why increasing the number of pipeline stages does not necessarily reduce the latency of an individual computation.

(d) **Hold-time consequences of pipelining.** **[5 points]**

For your optimal two-additional-register partition from part (b), evaluate the hold-time constraint at every newly created register-to-register stage.

Identify any stages that require additional minimum delay. Determine the minimum extra delay required for each violating stage.

Assume an inserted delay element increases both minimum and maximum delay by the same amount. Would these hold-time repairs change the minimum clock period you calculated in part (b)? Justify your answer.

Question 4: Time Borrowing in Latch-Based Pipelines [20 points]

A synchronous system uses alternating phases of level-sensitive latches, L1, L2, and L3. A latch is **ON** when it is transparent (data passes from input to output) and **OFF** when it holds its previous value.

L1 and L3 are controlled by phase ϕ_1, while L2 is controlled by phase ϕ_2.

For the timing interval under consideration:

- L1 turns ON at $t = 0$ ps and launches a new data value.
- L2 is ON from $t = 400$ ps to $t = 720$ ps.
- L3 is ON from $t = 800$ ps to $t = 1120$ ps.

Assume nonoverlapping clock phases. A latch is OFF outside its stated ON interval.

The launched data appears at the output of L1 65 ps after L1 turns ON.

The combinational logic between L1 and L2 has a maximum delay of 480 ps.

The combinational logic between L2 and L3 has a maximum delay of 420 ps.

L2 has a data-to-Q delay of 60 ps while it is ON. Both L2 and L3 require their input data to be stable at least 55 ps before the end of their ON intervals.

Assume no clock uncertainty and ignore minimum-delay constraints for parts (a)–(c).

(a) **Data propagation.** **[5 points]**

Determine when the launched data reaches:

1. The input of L2
2. The output of L2
3. The input of L3

Explain how L2 behaves when the data arrives while it is ON.

(b) **Time borrowing.** **[5 points]**

Determine whether the data satisfies the setup-time requirements of L2 and L3.

Calculate the setup timing margin at each latch.

How much time does the data borrow into L2's ON interval beyond the time at which L2 first turns ON?

(c) **Maximum allowable logic delay.** **[5 points]**

Keeping all other parameters unchanged, determine the maximum allowable delay of the combinational logic between L1 and L2 such that both L2 and L3 satisfy their setup-time requirements.

Now suppose the L1-to-L2 combinational delay increases by 60 ps from its original value. Determine which timing constraints are violated, if any.

(d) **Latches versus flip-flops.** **[5 points]**

Suppose L2 and L3 are replaced by positive-edge-triggered registers that capture data at $t = 400$ ps and $t = 800$ ps, respectively. Assume both registers have a setup time of 55 ps and a clock-to-Q delay of 60 ps, that the launch behavior of L1 is unchanged, and ignore minimum-delay constraints in this part as well.

Would the original logic delays satisfy the new timing requirements?

Explain why latch-based timing can permit a computation to complete even when an equivalent fixed-edge register boundary would fail.

Discuss one disadvantage of using latch-based timing in a large synchronous design.

Question 5: Race-Through, Clock Phasing, and Timing Robustness [20 points]

A designer is investigating a pipeline consisting of two positive-level-sensitive latches, L1 and L2, separated by combinational logic.

Initially, both latches are controlled by the same clock phase, which is high during the following intervals:

- 0–240 ps
- 600–840 ps

Both latches are opaque at all other times.

The input D of L1 is initially 0 and changes to 1 at $t = 20$ ps. It then remains stable through the first clock cycle.

The timing parameters are:

Parameter	Value
Transparent data-to-Q delay of L1	35 ps
Transparent data-to-Q delay of L2	40 ps
Minimum combinational logic delay	60 ps
Maximum combinational logic delay	190 ps
Setup time of L2	30 ps
Hold time of L2	25 ps

For a latch that is already transparent when its input changes, assume its output responds after the specified data-to-Q delay. If data becomes stable before the latch opens, assume its output responds after the same delay measured from the opening edge.

Assume ideal clock transitions, no clock uncertainty, and zero clock skew.

(a) **Same-phase latch behavior.** [5 points]

Determine the earliest and latest times at which the new data value can reach the input of L2.

For the minimum-delay path, determine the earliest time at which the output of L2 can change.

Does this allow the new input value to propagate through both latches during the same clock-high interval? Explain why this behavior is called race-through.

For the maximum-delay path, determine whether the input of L2 changes within its setup/hold timing window around the closing clock edge.

(b) **Fixing the race using delay insertion.** **[5 points]**

Suppose the designer wants to retain the same-phase clocking scheme but prevent new data from reaching L2 until its hold-time window after the first closing clock edge has ended.

Determine the minimum additional delay that must be introduced into the combinational path, assuming the added delay is the same for all paths.

Explain the effect of this change on the earliest and latest data arrival times at L2.

Discuss why inserting delay to resolve race-through can create other timing or performance concerns.

(c) **Two-phase latch design.** **[5 points]**

Instead of adding combinational delay, the designer changes the clocking scheme.

L1 remains transparent from 0 to 240 ps, but L2 is now transparent from 300 to 540 ps.

The original combinational logic delays remain unchanged.

Determine the earliest and latest times at which L2 receives the new data and when its output can change.

Does the new clocking scheme prevent race-through during the first clock-high phase? Does it satisfy the setup requirement of L2 before its new closing edge?

(d) **Timing-design recommendation.** **[5 points]**

Compare the following alternatives:

- Same-phase latches with additional delay inserted into the data path
- Two-phase latches with nonoverlapping transparency windows
- Positive-edge-triggered registers replacing both latches

Recommend an implementation for a high-performance synchronous system.

Your recommendation should discuss race-through, minimum-delay constraints, clock-phase generation, timing robustness, and opportunities for time borrowing.